Macroeconomic Determinants and Volatility Transmitter: Evidence from Conventional and Islamic Stock Market in Bangladesh

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Abstract

This study investigates the macroeconomic determinants of conventional and Islamic indices of Bangladesh based on macroeconomic version of the Arbitrage Pricing Theory (APT) and the semi-strong form of the Efficient Market Hypothesis (EMH) employing Autoregressive Distributed Lag (ARDL) bound testing method. Using monthly data from January 2014 to June 2017 of Dhaka Stock Exchange (DSE) Shariah index (DSES) from the Islamic stock market, DSE Broad index (DSEX) from the conventional stock market, broad money supply (M2), crude oil price (OP), exchange rate (ER), Bombay stock market index (SENSEX) and Dow Jones Islamic Market World (DJIMW) index, we find that conventional stock market in Bangladesh is effecient in semi-strong form of EMH. In contrast, results of the bounds test reveal that the long run cointegrations exist among the variables at 5% level of significance when DSES is dependent on M2, OP, ER, DSEX, SENSEX and DJIMW. The long run coefficients of ARDL results also show that DSEX have a positive and statistically significant long-run effect on DSES in Bangladesh, while M2 and OP are the two negative and significant determinants of DSES in long run. The coefficient of error correction term (-0.86) is negative and significant at 1% significance level suggesting that the long run causalities are also directing from M2, OP, ER, DSEX, SENSEX and DJIMW to DSE Shariah index. Results of the short run dynamics indicate that all of the variables have some significant impact on DSES. Specifically, in the short run, ER is the largest positive determinant of DSE Shariah index followed by DSEX, SENSEX and OP, while M2 and DJIMW are the two negative determinant of DSE Shariah index.

This paper also examines the volatility spillover effect using daily return data of DSES and DSEX over the period from 20 January 2017 to 30 June 2017. Results of a bivariate VAR model reveal the strong evidence of unidirectional spillovers in returns from conventional indices to the Islamic indices in Bangladesh. The volatility transmissions across markets are examined using MGARCH-BEKK model and results explore that the volatility is transmitted between these markets. Finally, employing MGARCH-CCC model, we find evidence of very high conditional correlation between Islamic and conventional indices of Bangladesh as the estimated conditional correlation (0.9) is significant at 1% significance level.

Key words: Arbitrage Pricing Theory (APT), Volatility Spillover, VAR, MGARCH-BEKK, MGARCH-CCC

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1. Introduction

Since Bangladesh has adapted free market economy as her national policy to achieve higher growth and rapid privatization, she must uplift her stock market and make it capable of working as the main vehicle for mobilizing and allocating funds needed to finance industrial and other development activities of the country. In fact, stock market can be considered as the creator of a strong and modern economy as it is more flexible and dynamic financial system than a traditional and rigid bank based financial systems. With an open market to foreign investors, stock markets of Bangladesh satisfies the definition of an emerging market in terms of income and market capitalization to GDP ratio. The equity market of Bangladesh has strengthened quickly over the years and highly contributed in 2010 about 50.67%. After that it was shown a downward trend and constituted about 18.40% of the country's gross domestic product in June, 2017 against only 7.59% of GDP in 2004 (DSE, 2017). When the trading activities restarted in 1976 after independence, Dhaka Stock Exchange (DSE) had only 9 listed companies with a paid up capital of Tk. 137.52 million. The number of listed securities has increased to 563 with a paid up capital of Tk. 3,788,581 million at the end of June 2017 (DSE, 2017). During the recent 15 years, the average economic growth rate has been approached to 6% per annum. Nominal GDP has been increased from Taka 50 billion to Taka 15,136 billion from fiscal year 1972-73 to 2014-15 (Hasan et al., 2016). Bangladesh is still a lower-middle income country as per capita income stands at around $1500. As Bangladesh aspires to be a high-income country by 2041, the stock market certainly needs to play more vital role accompanying with other sectors of the economy.

The increasing significance of stock markets globalization has amplified the interest in emerging markets. Accordingly, researchers have focused research on the efficiency and volatility dynamics of emerging stock markets as the issues contain various implications for investors and policymakers. The role of economic factors on the stock prices has been subjected to economic research all over the world. Harris and Pisedtasalasai (2006) argue that volatility spillover is an indicator of market efficiency. Thus, the analysis of financial market co-movement and volatility spillover between stock returns has been a central research agenda in economics and finance over the few years. But, the empirical studies on the determinants and volatility transmitter of conventional and Islamic indices in Bangladesh perspective are still scarce. Apart from the conventional stock markets, this study attempts to explore the efficiency and volatility dynamics in the context of Islamic equity markets. This is a major concern since Islamic finance is becoming one of the fastest growing segments of the global financial industry (Chiadmi and Ghaiti, 2014). The Islamic capital market progresses along with the conventional capital market and brings both Muslim and non-Muslim investors with an alternative investment values that is promptly mounting recognition. Islamic finance is not restricted only in Muslim countries rather spreads its Shariah practices to non-Muslim countries as several studies reveal that Islamic indices are better and outperform the conventional indices (Zamzamir et al., 2013). The global investors have also instigated seeking alternative assets to diversify their portfolio as the Islamic financial system was less affected during the recent global financial crisis. As a result, the global capital market has witnessed a torrent of Islamic indices over the past few years, such as, the Dow Jones Islamic Market (DJIM), the Standard & Poor Shariah Index (S&P), the Financial Times Islamic Index Series (FTSE), and the Morgan Stanley Capital International Islamic Index Series (MSCI), BSE 500 Shariah Index of Bombay Stock Exchange (Zamzamir et al., 2013). Although, Bangladesh is the first country in Southeast Asia where Islamic banking has been introduced in 1983, the first Islamic stock index in Bangladesh named Dhaka Stock Exchange (DSE) Shariah index (DSES) has been launched only just in January, 2014. Having a Muslim population of 90% and the fourth largest Muslim population in the World, Bangladesh has great prospect to develop a promising Islamic capital market.

Motivated from the above facts, the current study finds the answer to the following questions: i) Do the macroeconomic variables incorporated in this study share short run and long run relationships with the conventional or Islamic stock prices? ii) Who is the volatility transmitter between the conventional and Islamic stock indices in Bangladesh? iii) Is there any conditional correlations between the conventional and Islamic stock indices in Bangladesh? This study aims to contribute to the empirical literature as it explores the semi-strong form of efficiency of conventional and Islamic indices employing the latest cointegration and error correction model, named, ARDL bounds testing model. Moreover, the volatility dynamics of conventional and Islamic equity returns are examined employing GARCH family models. The organization of this study is as follows: section 2 focuses a brief literature review with theoretical framework; section 3 explains research methodology; section 4 presents empirical results; and section 5 concludes this study.

2. Theoretical Framework and Literature Review

The theories of economics and finance have constructively contributed to the real economy, and enriched the literature over the years. The existing theories of financial economics have provided a number of models explaining the link between stock market behavior and economic activity. Empirical studies have investigated the stock market behavior from the perspective of the Efficient Market Hypothesis (EMH) and the theory of asset pricing such as Portfolio Theory, Capital Asset Pricing Model (CAPM), Present Value Model (PVM) and Arbitrage Pricing Theory (APT). The theoretical basis of this research is consistent with the EMH and the most prominent equilibrium asset pricing model named APT. The EMH is probably the most dominant and well-known market theory in the world. Usually, the empirical efforts of testing stock market efficiency emphasis on the informational market efficiency. The concept of efficient capital market has not emerged all of a sudden in the 1960s. It has slowly evaluated from the independent works of several economists, and these contributions have succeeded by Fama's (1965) definition of EMH. In the same year, Paul A. Samuelson provides the first formal economic argument for efficient markets focusing on the concept of a martingale, while Fama defines an efficient market for the first time and concludes that the stock prices follow a random walk. The EMH states that stock market prices already incorporate and reflect all available information such that none can beat the market by making abnormal profits (Fama, 1965). A stock market is said to be efficient if stock prices in that market reflect all available information. Fama (1970) mentions that EMH exists in various degrees: weak, semi-strong, strong and successively he accomplishes an important contribution in making the efficient market hypothesis testable. The weak form of EMH assumes that past stock prices are not significant to make an abnormal return. The semi-strong form of EMH assumes that no investor can earn abnormal returns using publicly available information. Finally, the strong form of EMH assumes that all public and private information are reflected by present stock prices. In practice, there are four different methods exercised to test the EMH. The methods are technical analysis, fundamental analysis, macroeconomic analysis, and mutual funds and insider trading behavior analysis. Fundamental analysis and macroeconomic analysis are related to test the semi-strong form efficiency of the EMH. Fundamental analysis is the microeconomic version and uses all publicly available firm-specific information including historical data of stock prices to test the semi strong form efficiency. Macroeconomic analysis uses all publicly available macroeconomic information including historical data of stock prices. In thid analysis, a stock market is said to be efficient in semi strong form if no investor can earn abnormal profit using publicly available global and domestic macroeconomic information such as GDP, interest rate, exchange rate, money supply, foreign stock markets index etc. The Arbitrage Pricing Theory (APT) is developed by Ross (1976) that expands the original CAPM from a single-factor to a multi-factor model. APT states that a number of systematic risk factors that are generated by the variations of macroeconomic and financial variables which determine the market return. Two versions are used to test and implement APT. Factor loading version uses artificial variables related statistical (i.e., principal component analysis, factor analysis) techniques, while macro variable version uses economic factors. The APT is the most popular asset pricing model that is empirically investigated by a significant number of researchers as it grants the freedom to select variables to the economy. The systematic risk factors may be economic factors such as interest rates, inflation, GDP, and financial factors such as market indices, yield curves, exchange rates. Factors may also be fundamentals like price-earnings ratios, dividend yields, etc.

Leigh (1997) examines the three forms of EMH in the stock market of Singapore, and the relationship between the stock market and the overall economy using quarterly data from 1975 Q1 to 1991 Q2. Employing unit root tests, he shows that the stock returns of Singapore stock market follow random walk model, therefore the market is efficient in weak form. Based on Johansen-Juselius Cointegration model, the study finds that there is no cointegration between stock prices and macroeconomic variables. Thus, he comments that the stock market follows semi strong form of EMH. Moreover, the study operates ECM and finds that stock market of Singapore does not follow the strong form of EMH. The Granger causality test suggests that Singapore stock market index can be used as a leading indicator of the economy. Finally, ARCH and GARCH model are used to test the excess volatility, and the results find no significant evidence of asset pricing inefficiency in Singapore stock market. **Chen, Roll, and Ross (1986)** test seven macroeconomic variables, namely, term structure, industrial production, risk premium, inflation, market return, consumption, and oil prices in the period of January 1953 to November 1984 to explain the U.S stock return. They reveal that several of these economic variables are significant in explaining expected stock return. **Isenmila and Erah (2012)** examine the appropriateness of the APT to give reasons for stock returns in Nigeria for the period 2000 Q1-2010 Q4. The study examines the implication of oil prices (OP), money supply (M2), Gross Domestic Product, Exchange rate (ER), inflation, and interest rate (IR) in explaining stock returns (All share index) of the Nigerian stock market. The cointegration and error correction methodology reveal that M2 and OP are emerged to be negative significant determinant, while ER and IR are observed to be negative insignificant determinant of stock returns both in the long run and the short run. They recommend that the APT macroeconomic variables can explain stock returns and thus there is the need for practical harmonization of macroeconomic policies in Nigeria. **Rayhan, Sarkar and Sayem (2012)** examine the volatility of DSE returns using unit root tests and GARCH model for the period of January 1987 to March 2010. They comment that DSE price index follows random walk of EMH, but monthly DSE returns do not follow it. They also reveal that the stock market volatility changes significantly over time and monthly DSE returns follow GARCH properties. At the aggregate level, stock return volatility follows leverage effect or asymmetric volatility as the study uncovers that volatility rises more sharply during stock price declines following bad news than that in periods of stock price increase following good news. **Gahlot (2013)** intends to study volatility spillover among South Asian Countries (Bangladesh, India, Pakistan and Sri Lanka) and U.S. stock markets through use of Granger causality test and C GARCH M. He finds significant bidirectional causality between Stock market of U.S. and India for both short terms as well as for long term which is not disturbed by recession. But the recession has changed causal relation among other countries. The study also reveals that volatility of all South Asian countries is of long term nature and the observed spillover effects are unstable over time in sense that the spillover changed its nature after beginning of recession. Using the dataset over the period from 2000 to 2011 and covering three major regions: Europe, the USA, and the world, **Jawadi, Jawadi, and Louhichi (2014**) find that Islamic indices outperformed their conventional peers during financial crisis period. They extend utilizing CAPM-GARCH to correct the bias while it captures volatility dynamics. **Chiadmi and Ghaiti (2014)** investigate the volatility behavior of the Standard and Poor's Sharia'h index (S&P Sharia'h), the Dow Jones Islamic Market (DJIM) index, The FTSE Islamic index, The MSCI Islamic World as well as their conventional counterparts, respectively, the S&P 500, the Dow Jones Industrial Average (DJIA), the FTSE All world and the MSCI World Indexes. Results of the GARCH family models reveal that Islamic stock indexes were significantly affected by the financial crisis but they were less volatile than their conventional counterparts. **Hasan and Wadud (2015)** investigates semi-strong form of the Efficient Market Hypothesis (EMH) based on macroeconomic variable version of the Arbitrage Pricing Theory (APT) using monthly data of All Share Price Index (DSI) of Dhaka Stock Exchange (DSE) and five macroeconomic variables namely, Industrial Production Index (IPI), Broad Money Supply (M2), Crude Oil Price (OP), Exchange Rate (ER) and Index of Bombay Stock Exchange (SENSEX) from January 2001 to December 2012. The Johansen and Juselius multivariate cointegration tests reveal that IPI and OP have significant negative long run relationship with DSI, while M2, ER and SENSEX have significant positive long run relationship with DSI. The study also reveals that individually IPI and SENSEX are the leading indicators with respect to stock prices in Bangladesh in the short run. Moreover, stock price index of DSE is a leading indicator with respect to IPI and ER in the short run. Hence, they conclud that DSE is not efficient in the semi-strong form of EMH. **Mohammadi and Tan (2015)** examines the dynamics of daily returns and volatility in stock markets of the U.S., Hong Kong and mainland China (Shanghai and Shenzhen) over 2 January 2001 to 8 February 2013 employing VAR and MGARCH models. The results suggest the evidence of unidirectional return spillovers from the U.S. to the other three markets; but no spillover between Hong Kong and either of the two mainland China markets. The study also finds the evidence of unidirectional ARCH and GARCH effects from the U.S. to the other three markets. The patterns of dynamic conditional correlations from the DCC model suggest an increase in correlation between China and other stock markets since the most recent financial crisis of 2007. **Henda and Taher (2017)** investigates Islamic Versus Conventional market indexes' performance using global data of four Islamic indices and their four conventional counterparts from 2003 to 2011. The findings provide evidence that index performance are somewhat mixed over the different period and through the different indices under consideration, and support the hypothesis that the impact of faith-based screens on investment performance is insignificant. They also find that there is no long run relationship between the Islamic indices and their conventional counterparts' performance, except for the Islamic emerging markets indices over the three sub-periods. Finally, in the short-run, they find different causal links between Islamic Versus non-Islamic indices over the three sub-periods.

3. Methodology

3.1 Data and Data Sources

In order to find the determinants of the stock prices, a multifactor model is formulated with the help of macroeconomic version of the Arbitrage Pricing Theory (APT) and the semi-strong form of the Efficient market Hypothesis (EMH). For the multivariate models, we use monthly log data from January 2014 to June 2017 of DSE Shariah index (DSES) from the Islamic stock market, DSE Broad index (DSEX) from the conventional stock market, broad money supply (M2) from the money market, crude oil price (OP) from the natural resources market, exchange rate (ER) from the foreign exchange, Bombay stock market index (SENSEX) from the foreign market and Dow Jones Islamic Market World (DJIMW) index are considered from the foreign Islamic market. This paper also examines the volatility dynamics using daily return data of DSES, DSEX and DJIMW over the period from 20 January 2017 to 30 June 2017. The data are collected from secondary sources, such as, the official website of Dhaka Stock Exchange, Monthly Economic Trend issued by Bangladesh Bank, official website of Bombay Stock Exchange and official website of S&P Dow Jones Indices. The analysis is done using the EViews 9 econometric software packages.

3.2 Research Methods

3.2.1 Unit Root Test

Two widely used unit root test, namely Augmented Dickey Fuller (ADF) and Phillips-Peron (PP) test are employed to examine the order of integration of the time series. ADF is the augmented form of Dickey Fuller test. The ADF test is performed using the following equation:

$$\Delta Y_t = \alpha + \beta T + \gamma Y_{t-1} + \delta_i \sum_{i=1}^{m} \Delta Y_{t-i} + \varepsilon_t \qquad (1)$$

where, α is a interecpt (constant), β is the coefficient of time trend T, γ and δ are the parameters where , $\gamma = \rho$-1, ΔY is the first difference of Y series, m is the number of lagged first differenced term, and ε is the error term.

Phillips and Perron (1988) have developed a non-parametric unit root conception. The PP test test is performed using the following three equations:

$$\Delta Y_t = \alpha + \beta T + \gamma Y_{t-1} + \varepsilon_t \qquad (2)$$

where, α is a constant, β is the coefficient of time trend T, γ is the parameter and ε is the error term.

3.2.2 ARDL Bounds Test

This study uses Autoregressive Distributed Lag (ARDL) modeling to cointegration procedure of Pesaran, Shin and Smith (2001). ARDL model has few advantages in comparison to the conventional cointegration procedures, such as, residual-based approach proposed by Engle and Granger (1987) and the maximum likelihood approach proposed by Johansen and Juselius (1990). First, ARDL model can be applied on a time series data irrespective of whether the variables are I(0) or I(1), while Johansen cointegration techniques require that all the variables in the system be of equal order of integration (Pesaran and Pesaran, 1997). Second, the ARDL procedure permits that the variables may have different optimal lags, while it is impossible with conventional cointegration procedures. Third, the ARDL procedure is statistically more significant approach to determine the cointegration relation in small samples, while the Johansen cointegration procedures need large data samples for validity (Pesaran, Shin and Smith, 2001; Haug, 2002). Fourth, the ECM can be derived from ARDL integrates the short-run dynamics with the long run equilibrium without losing long-run information. Finally, the ARDL procedure makes use of only a single reduced form equation, while the conventional cointegration procedures estimate the long-run relationships within a context of system of equations.

The ARDL long-run models can be expressed mathematically as:

$$DSES_t = \alpha_1 + \beta_1 M2_t + \beta_2 OP_t + \beta_3 ER_t + \beta_4 DSEX_t + \beta_5 SENSEX_t + \beta_6 DJIMW_t + \varepsilon_{1t} \quad (3)$$

$$DSEX_t = \alpha_1 + \beta_1 M2_t + \beta_2 OP_t + \beta_3 ER_t + \beta_4 DSES_t + \beta_5 SENSEX_t + \beta_6 DJIMW_t + \varepsilon_{1t} \quad (4)$$

where α, β and ε represent constants, coefficients and error terms respectively. Equation (3) and (4) can be re-expressed in the following conditional error correction model (ECM) version of the ARDL to implement the bounds testing procedure:

$$\begin{aligned}\Delta DSES_t = {} & c_1 + \pi_1 DSES_{t-1} + \pi_2 M2_{t-1} + \pi_3 OP_{t-1} + \pi_4 ER_{t-1} + \pi_5 DSEX_{t-1} + \pi_6 SENSEX_{t-1} \\ & + \pi_7 DJIMW_{t-1} + \sum_{i=1}^{\rho} \theta_i \, \Delta DSES_{t-i} + \sum_{i=1}^{\rho} \emptyset_i \, \Delta M2_{t-i} + \sum_{i=1}^{\rho} \delta_i \, \Delta OP_{t-i} \\ & + \sum_{i=1}^{\rho} \gamma_i \, \Delta ER_{t-i} + \sum_{i=1}^{\rho} \eta_i \, \Delta DSEX_{t-i} + \sum_{i=1}^{\rho} \psi_i \, \Delta SENSEX_{t-i} + \sum_{i=1}^{\rho} \omega_i \, \Delta SENSEX_{t-i} \\ & + u_{1t} \qquad (5)\end{aligned}$$

$$\begin{aligned}\Delta DSEX_t = {} & c_1 + \pi_1 DSEX_{t-1} + \pi_2 M2_{t-1} + \pi_3 OP_{t-1} + \pi_4 ER_{t-1} + \pi_5 DSES_{t-1} + \pi_6 SENSEX_{t-1} \\ & + \pi_7 DJIMW_{t-1} + \sum_{i=1}^{\rho} \theta_i \, \Delta DSEX_{t-i} + \sum_{i=1}^{\rho} \emptyset_i \, \Delta M2_{t-i} + \sum_{i=1}^{\rho} \delta_i \, \Delta OP_{t-i} \\ & + \sum_{i=1}^{\rho} \gamma_i \, \Delta ER_{t-i} + \sum_{i=1}^{\rho} \eta_i \, \Delta DSES_{t-i} + \sum_{i=1}^{\rho} \psi_i \, \Delta SENSEX_{t-i} + \sum_{i=1}^{\rho} \omega_i \, \Delta SENSEX_{t-i} \\ & + u_{1t} \qquad (6)\end{aligned}$$

where (5) and (6) are termed as model 1 and 2 respectively. The first parts of the equations represent the long-run dynamics of the models and the second parts show the short-run relationship in which Δ signifies the first difference operator. c_i ($i = 1, 2$) show constants, π_i ($i = 1..6$) denote coefficients on the lagged levels, θ_i, ϕ_i, δ_i, γ_i, η_i, Ψ_i and ω_i ($i = 1 \ldots \rho$) denote coefficients on the lagged variables, and finally u_i($i = 1 \ldots 6$) stand for error terms. ρ signifies the maximum lag length, which is decided by the Akaike Information Criterion (AIC). The Schwartz Bayesian Information Criterion (SBC) and Akaike Information Criterion (AIC) are two lag length criteria popularly used to select the maximum lag length in autoregressive models. The AIC is utilized in this study to determine the order of the ARDL model as it has a lower prediction error than that of the SBC (Hasan et al. 2017).

Pesaran, Shin and Smith (2001) argues that the ARDL bounds testing method consists of two steps. The first step inspects the presence of long-run relationship, while the second step estimates the long run and short-run coefficients of the models. Thus, we estimate the equations (5) and (6) in order to test the long-run relationship by conducting F-test for the joint significance of the coefficients of the lagged levels of the variables. The null and alternative hypotheses are as follows:

$H_0: \pi_1 = \pi_2 = \pi_3 = \pi_4 = \pi_5 = \pi_6 = 0$ (No long run relationship)

$H_1: \pi_1 \neq \pi_2 \neq \pi_3 \neq \pi_4 \neq \pi_5 \neq \pi_6 \neq 0$ Long run relationship exists)

Pesaran, Shin and Smith (2001) states that two sets of critical values for a given significance level can be determined. The first level is calculated on the assumption that all variables incorporated in the ARDL model are integrated of order zero, while the second one is calculated on the assumption that the variables are integrated of order one. The null hypothesis of 'no cointegration' is rejected when the value of the test statistic exceeds the upper critical bounds value, while it is accepted if the F-statistic is lower than the lower bounds value. Otherwise, the cointegration test is inconclusive.

Finally, we perform diagnostic and stability tests to establish the goodness of fit for the ARDL models. The diagnostic tests include the serial correlation, normality and heteroscedasticity associated with the models. The stability tests are conducted by operating the cumulative sum of recursive residuals and the cumulative sum of squares of recursive residuals.

3.2.3 Vector Autoregressive (VAR) Model

This study examines the possibility of spillover in conventional and Islamic stock returns in Bangladesh using VAR-Granger Causality model. Vector Autoregression (VAR) is used to find the relationship between various variables by taking into account the feedback by other variables. Estimation based on VAR contains endogenous variables in adding exogenous variables in the equation, making it an important tool for multivariate analysis. As in Enders (2004), an n-equation VAR can be represented by:

$$\begin{bmatrix} X_{1t} \\ X_{2t} \\ . \\ X_{nt} \end{bmatrix} = \begin{bmatrix} A_{10} \\ A_{20} \\ . \\ A_{n0} \end{bmatrix} + \begin{bmatrix} A_{11}(L) & A_{12}(L) & . & A_{1n}(L) \\ A_{21}(L) & A_{22}(L) & . & A_{2n}(L) \\ . & . & . & . \\ A_{n1}(L) & A_{n2}(L) & . & A_{nn}(L) \end{bmatrix} \begin{bmatrix} X_{1t-1} \\ X_{2t-1} \\ . \\ X_{nt-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ . \\ \varepsilon_{nt} \end{bmatrix} \quad (7)$$

where A_{i0} = the parameters representing intercept terms, $A_{ij}(L)$ = the polynomials in the lag operator L. The individual coefficients of $A_{ij}(L)$ are denoted by $a_{ij}(1)$, $a_{ij}(2)$,... Since all equations have the same lag length, all the polynomials $A_{ij}(L)$ are of the same degree. The terms ε_{it} are white-noise disturbances that may be correlated. Again designate the variance/covariance matrix by $\sum$, where the dimension of $\sum$ is (n xn).

This study examines the possibility of spillovers in returns over time and across market using a two-variable VAR model consisting of returns for DSES and DSEX. A two-equation VAR can be represented by where the returns of Islamic market depend on *p* lags of itself, as well as *p* lags of the returns of conventional market:

$$\mathrm{DSES_t} = \varphi_1 + \sum_{i=1}^{\rho} \varphi_{\mathrm{DSESi}}\, \mathrm{DSES}_{,t-i} + \sum_{i=1}^{\rho} \varphi_{\mathrm{DSEXi}}\, \mathrm{DSEX}_{,t-1} + \varepsilon_{1,t} \tag{8}$$

The possibility of spillover in returns over time can be examined by testing the joint hypotheses that $\varphi_{DSESi} = 0$ ($i = 1, \ldots, p$). Equally, the possibility of spillovers in returns from conventional market to Islamic market can be examined by testing for the joint hypotheses that $\varphi_{DSEXi} = 0$ ($i = 1, \ldots, p$).

3.2.4 MGARCH-BEKK

In order to capture the co-movement volatility between conventional and Islamic stock returns in Bangladesh, this study uses multivariate GARCH-BEKK (Baba-Engle-Kraft-Kroner) model. MGARCH-BEKK model is an extended version of the GARCH model which can capture volatility transmission among different series as well as the persistence of volatility within each series. BEKK formulation enables us to reveal the existence of any transmission of volatility from one market to another (Engle and Kroner, 1995). The BEKK model of Engle and Kroner (1995) can be written as:

$$\mathrm{H_t} = \mathrm{CC'} + \sum_{i=1}^{k} \mathrm{A_i}\, \varepsilon_{t-1}\varepsilon'_{t-i}\mathrm{A'_i} + \sum_{i=1}^{k} \mathrm{B_i H_{t-i}\, B'_i} \tag{9}$$

where C, A_i and B_i are N×N matrics, but C is triangular. This equation guarantees all positive definite diagonal representation.

To illustrate the BEKK model, consider the simple GARCH (1,1) model:

$$\mathrm{H_t} = \mathrm{CC'} + A_1 \varepsilon_{t-1}\varepsilon'_{t-1}\mathrm{A'_1} + \mathrm{B_1 H_{t-1} B'_1} \tag{10}$$

In the bivariate case as in this study, the BEKK becomes:

$$\mathrm{H_t} = \mathrm{CC'} + \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} \varepsilon^2_{1,t-1} & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{2,t-1}\varepsilon_{1,t-1} & \varepsilon^2_{2,t-1} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}'$$
$$+ \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} h_{11,t-1} & h_{12,t-1} \\ h_{21,t-1} & h_{22,t-1} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}' \tag{11}$$

where the symmetric matrixes A captures the ARCH effects, matrixes B focus on the GARCH effects. The diagonal parameters in matrixes A and B measure the effects of own past shocks and past volatility on its conditional variance. The off-diagonal parameters in matrixes A and B, a_{ij} and b_{ij} measure the cross-market effects of shock and volatility; also known as volatility spillover.

In the BEKK model, the ARCH component associated with the conditional variance of DSEX can be written as:

$$h_{11,t} = C_1 + a_{11}^2\varepsilon_{1,t-1}^2 + a_{21}^2\varepsilon_{2,t-1}^2 + 2a_{11}a_{21}\varepsilon_{1,t-1}\varepsilon_{2,t-1} \quad (12)$$

where the ARCH volatility in the DSEX return depends on the squares as well as the cross products of the previous shocks associated with the two markets. Here, a_{11} and a_{21} capture the effects of past squared shocks in each market on today's volatility in DSES returns.

Similarly, The GARCH component of the DSES conditional variance can be written as:

$$h_{11,t} = b_{11}^2 h_{11,t-1} + b_{21}^2 h_{22,t-1} + 2b_{11}b_{21}h_{12,t-1} \quad (13)$$

where the volatility of DSES return depends on the past conditional variances and covariances associated with each markets of the two markets. Here, b_{11} and b_{21} capture the effects of past volatility in each of the two markets on today's volatility in DSES.

3.2.5 MGARCH-CCC

Constant Conditional Correlation (CCC) model is developed by Bollerslev (1990). This model assumes that correlations between each pair of returns are constant and thus the volatility model consists only of the equations for the variances. CCC model has been very popular among empirical studies because it reduces the conditional correlation matrix to constant correlation coefficients between variables, so the number of parameters to be estimated is small in comparison with other models. The conditional covariance matrix is defined as:

$$H_t = S_t R S_t \quad (14)$$

where S_t is a (N×N) diagonal matrix of time-varying standard deviations and R is a (N×N) matrix of constant correlations.

In the bivariate case as in this study, the CCC becomes as follows:

$$H_t = \begin{bmatrix} \sqrt{h_{11,t}} & 0 \\ 0 & \sqrt{h_{22,t}} \end{bmatrix} \begin{bmatrix} 1 & \rho_{12} \\ \rho_{21} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{h_{11,t}} & 0 \\ 0 & \sqrt{h_{22,t}} \end{bmatrix} \quad (15)$$

In this case, H_t is assumed to be positive definite if certain restrictions on the parameters are correctly satisfied. Variance terms $h_{11,t}$ and $h_{22,t}$ are univariate GARCH processes with p=q=1.

4. Empirical Results

4.1 Unit Root Test Results

Results of the ADF and Phillips-Perron unit root tests given in Table 1 show that the variables are nonstationary in levels. Results also show that all series are stationary in first differences with 1% significance level. Since none of the variable are intregrated of order two, i.e., I (2), we can proceed our study applying the ARDL bound testing method.

Table 1. Results of ADF and PP Unit Root Tests

Variables	ADF		PP	
	Intercept	Trend & Intercept	Intercept	Trend & Intercept
DSES	-1.87 (0.34)	-2.60 (0.28)	-1.78 (0.38)	-2.62 (0.27)
ΔDSES	-6.93*(0.00)	-6.84* (0.00)	-7.64* (0.00)	-7.53* (0.00)
DSEX	-1.20 (0.66)	-1.85 (0.66)	-1.20 (0.66)	-1.85 (0.66)
ΔDSEX	-6.53 (0.00)	-6.65 (0.00)	-6.54* (0.00)	-6.81* (0.00)
M2	-1.13 (0.69)	-2.39 (0.38)	-1.12 (0.66)	-2.57 (0.30)
ΔM2	-7.13* (0.00)	-7.19* (0.00)	-7.14* (0.00)	-7.19* (0.00)
OP	-1.84 (0.36)	-0.99 (0.93)	-1.59 (0.48)	-0.99(0.93)
ΔOP	-4.54* (0.00)	-4.65* (0.00)	-4.52* (0.00)	-4.47* (0.00)
ER	2.38 (0.99)	-0.09 (0.99)	-2.11 (0.99)	-2.34 (0.99)
ΔER	-4.35* (0.00)	-4.95* (0.00)	-4.31* (0.00)	-4.96* (0.00)
SENSEX	-2.58 (0.10)	-2.51 (0.32)	-2.58 (0.10)	-2.54 (0.31)
ΔSENSEX	-5.94* (0.00)	-5.93* (0.00)	-5.94* (0.00)	-5.93* (0.00)
DJIMW	-1.67 (0.44)	-2.07 (0.54)	-1.73 (0.41)	-2.07 (0.55)
ΔDJIMW	-6.81* (0.00)	-6.81* (0.00)	-7.20* (0.00)	-7.72* (0.00)

Note: First bracket shows p values. * indicates stationary at 1% level using MacKinnon (1996) critical and p values.

4.2 Results of ARDL Bounds Testing Approach

The ARDL procedure estimates the models and tests for the presence of cointegration among the variables. The AIC selects the ARDL (4, 4, 4, 4, 3, 4, 4) for the variables included in the model 1, while the AIC chooses the ARDL (4, 4, 4, 4, 4, 4, 4) for the model 2. The result of the F-Statistic is presented in Table 2 which shows that the computed F- statistics is 3.72 for model 1 that is higher than the upper bound critical value of 3.28 at 5% level of significance. Therefore, the long-run relationship among the variables exists when Islamic equity indice is dependent on M2, OP, ER, DSEX, SENSEX and DJIMW. On the other hand, the long-run relationship is not evident when conventional equity indice is dependent on domestic and global macroeconomic variables at 5% level of significance.

Table 2. Results of ARDL Bounds Cointegration Test

Dependent Variable	Forcing Variables	F-Statistics	5% Critical Bounds		Remarks
			I(0)	I(1)	
1: DSES	M2, OP, ER, DSEX, SENSEX, DJIMW	3.72*	2.27	3.28	Cointegration Presents
2: DSEX	M2, OP, ER, DSES, SENSEX, DJIMW	3.04	2.27	3.28	No Cointegration

Note: * denotes rejection of the null hypothesis at the 5% level.

Table 3 illustrates the long run coefficients of DSES (Islamic equity indice). The ARDL results show that DSEX is positively related with DSES, while M2 and OP are negatively related. The result implies that a 1% increase in conventional indice contributes to 0.63 % increase in Islamic equity indice in Bangladesh. Besides, a 1% increase in broad money supply and crude oil price contributes to 01.03% and 0.26% decrease respectively in Islamic equity indice. Thus, we can state that DSEX, M2 and OP have some significant impact on DSES in the long run.

Table 3. Long-run Coefficients for DSES

Variable	Coefficient	Std. Error	t-Statistic	Prob.
M2	-1.031221**	0.435052	-2.370340	0.0768
OP	-0.256336**	0.109312	-2.344987	0.0789
ER	4.164832	2.379759	1.750107	0.1550
DSEX	0.630053*	0.199318	3.161047	0.0341
SENSEX	0.181671	0.201269	0.902626	0.4178
DJIMW	0.701612	0.706144	0.993581	0.3767
C	-13.571815	7.973192	-1.702181	0.1639

Note: * and ** denote significance at 5% and 10% levels respectively.

The short-run dynamics along with the error correction term (ECT) results are reported in Table 4. The results of the estimated ARDL-ECM models clearly indicate that the coefficients of error correction terms of the model is negative and statistically significant at the 1% level of significance. It suggests that the long-run causality is also directing from independent variables to DSES. The error correction term of model 1 is -0.86, which implies that the system requires about 9 months to converge to equilibrium after being shocked. The short-run significant impacts of OP, ER, DSEX and SENSEX on DSES are positive, while the roles of M2 and DJIMW are negative.

Table 4. Error Correction Estimates

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(DSES(-1))	-0.491611*	0.088911	-5.529227	0.0052
D(M2)	-1.136925*	0.171143	-6.643132	0.0027
D(OP)	0.043967**	0.015400	2.855020	0.0462
D(ER)	6.337230*	0.926547	6.839622	0.0024
D(DSEX)	1.136147*	0.028461	39.919610	0.0000
D(SENSEX)	0.255014*	0.049028	5.201372	0.0065
D(DJIMW)	-0.202363**	0.069250	-2.922194	0.0431
CointEq(-1)	-0.860330*	0.095041	-9.052187	0.0008

Note: *and**denote significance at 1% and 5% levels respectively.

Thus, we can comment that the conventional indice in Bangladesh is efficient in semi-strong form based on EMH and APT, while the Islamic indice is not efficient in semi-strong form. In the long run, conventional equity indice is the most positive determinant of Islamic equity indice in Bangladesh, while exchange rate in the short run.

In order to verify the robustness of results, diagnostic checking of the estimated model has been carried out in terms of conventional multivariate residual-based tests for serial correlation, normality and heteroscedasticity and results are given in Table 5. At 1% level of significance, the Lagrange Multiplier (LM) test for autocorrelation indicates the absence of autocorrelation and

ARCH Chi-square test for heteroskedasticity indicates the absence of heteroskedasticity. The model also passes the Jarque-Bera normality test at 1 percent suggesting that the error is normally distributed in the models.

Table 5. Results of Diagnostic Tests

Diagnostic Tests	F-Statistics	P-Value
Serial Correlation LM	9.08	0.10
ARCH Heteroskedasticity	0.61	0.55
Jarque-Bera Normality	0.40	0.82

Finally, the cumulative sum (CUSUM) of recursive residuals and the cumulative sum of squares (CUSUMSQ) of the recursive residuals tests are employed to assess parameter stability. Figure 1 plots the statistics of CUSUM and CUSUMSQ of recursive residuals. The plotted points for the CUSUM and CUSUMSQ statistics stay within the critical bounds of 5% level of significance. Thus, these statistics confirm the stability for all coefficients of the estimated model.

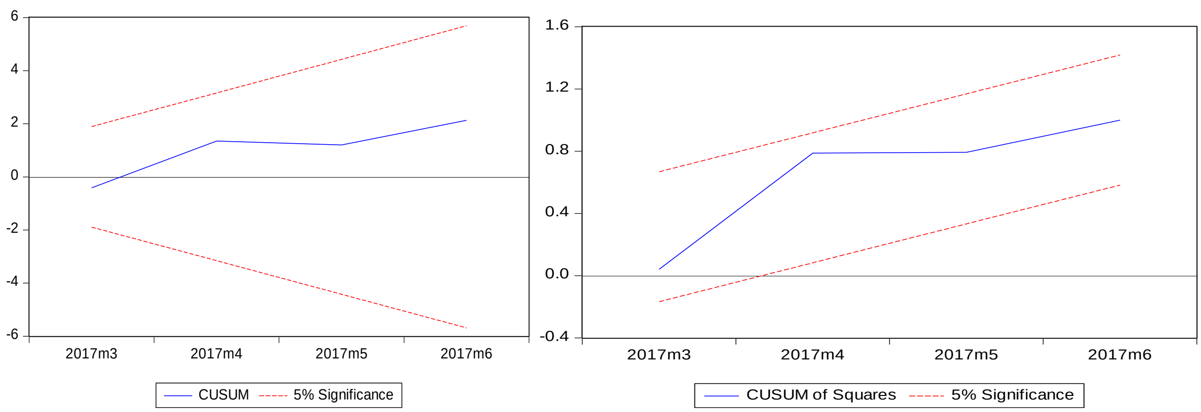

Figure 1. Plots of CUSUM and CUSUMSQ Stability Test

4.3 VAR-Granger Causality Results

This study uses and checks five different criteria namely, Likelihood Ratio (LR), Final Prediction Error (FPE), Akaike Information Criteria (AIC), Schwarz Information Criteria (SIC) and Hannan-Quinn Information Criteria (HQ) to determine the optimum lag lengths of the VAR model. Results for each criterion with a maximum of 7 lags exhibit that AIC, sequential modified LR and FPE criteria stand in favor of 5 lags ,while SIC and HQ criteria suggest for only 1 lag. The presence of residual serial correlation makes the result less efficient. Thus, we proceed to conduct LM tests for each suggested lags up to maximum 5 lags. Using 7 lags suggested by LR, AIC and FPE criterion produce no autocorrelation in the VAR model for up to 7days at 1% significance level. So, we accept VAR (5) model for Granger Causality/Block Exogeneity Wald tests between two markets.

The reported results in Table 6 reveal that there is evidence of unidirectional spillovers in returns from the conventional stock market to Islamic stock market, while there is no evidence of causality from DSES to DSEX.

Table 6. Estimates of VAR Granger Causality Test

Dependent	Excluded	Chi-sq	df	Prob.
RDSEX	RDSES	7.40	5	0.19
RDSES	RDSEX	11.10*	5	0.04

Note: * indicates significance at the 5% level.

4.4 MGARCH-BEKK Results

Panel A of the table 7 reports the estimates of BEKK parameters. Here, the ARCH effects reveal that the diagonal elements A(1, 1) and A(2, 2), are statistically significant, signifying that each conditional variance depends on its own squared lagged innovations. The DSES has the largest own ARCH effect with the coefficient value of 0.328697 and there is evidence of a bidirectional ARCH effect between DSES and DSEX. The GARCH effects reveal that two conditional variances depend on their own history; this is evident in the estimated B(1, 1) and B(2, 2) GARCH parameters, which are all significant at the 1% level. The results also show that the DSEX has the largest own GARCH effect and there is evidence of a bidirectional volatility spillover between coventional and Islamic stock markets. Panel B of the table reports two diagnostic tests for each market. We fail to reject the null hypothesis and these results provide some indications of misspecification in the GARCH-BEKK model.

Table 7. Estimates of ARCH and GARCH parameters in the BEKK model

	DSES/DSEX			
	Panel A. Parameter Estimates			
	Coefficient	Std. Error	z-Statistic	Prob.
M(1,1)	0.011283	0.003514	3.211170	0.0013
M(1,2)	0.008527	0.002537	3.360997	0.0008
M(2,2)	0.007921	0.002220	3.568978	0.0004
A1(1,1)	0.328697	0.021170	15.52646	0.0000
A1(2,2)	0.307058	0.021238	14.45782	0.0000
B1(1,1)	0.931017	0.008938	104.1605	0.0000
B1(2,2)	0.942600	0.007326	128.6575	0.0000
	Panel B. Diagnostic Tests (Portmanteau Tests for Autocorrelations)			
Q(30)	222.82 (0.00)			
Adj Q(30)	225.44 (0.00)			

Notes: A and B are the corresponding ARCH and GARCH parameters for markets.

4.5 MGARCH-CCC Results

The performance of the MGARCH-CCC model is reported in Table 8. This model estimates the own ARCH and GARCH effect as well as the correlations between each of the two markets. The results suggest the existence of own ARCH and GARCH effects in both markets as all of the estimated parameters are significantly different from zero and significant at 1% level. The estimated conditional correlation is positive and significantly different from zero at 1% significance level. The high conditional correlation (0.90) between DSES and DSEX reflect the presence of strong direct interconnections between coventional and Islamic stock markets in Bangladesh. Finally, according to the diagnostics for the CCC model, we fail to reject the null hypothesis of serial independence in the DSES and DSEX at 30 lags.

Table 8. **Estimates of the** MGARCH-CCC model

	Transformed Variance Coefficients			
	Coefficient	Std. Error	z-Statistic	Prob.
M(1)	0.007318	0.002167	3.377338	0.0007
A1(1)	0.050494	0.006325	7.982776	0.0000
B1(1)	0.929175	0.009021	102.9980	0.0000
M(2)	0.003213	0.001035	3.104142	0.0019
A1(2)	0.032533	0.005278	6.163448	0.0000
B1(2)	0.957237	0.006824	140.2822	0.0000
R(1,2)	0.900716	0.006086	147.9936	0.0000
	Diagnostic Tests (Portmanteau Tests for Autocorrelations)			
Lags	Q-Stat	Prob.	Adj Q-Stat	Prob.
30	222.82	0.0000	225.44	0.0000

5. Conclusion

This study explores the semi-strong form of efficiency of conventional and Islamic indices employing the ARDL bounds testing model using monthly data from January 2014 to June 2017 of Dhaka Stock Exchange (DSE) Shariah index (DSES) from the Islamic stock market, DSE Broad index (DSEX) from the conventional stock market, broad money supply (M2), crude oil price (OP), exchange rate (ER), Bombay stock market index (SENSEX) and Dow Jones Islamic Market World (DJIMW) index. Moreover, the volatility dynamics of conventional and Islamic equity returns are examined employing VAR, MGARCH-BEKK and MGARCH-CCC models. In doing so, the study finds that conventional stock market in Bangladesh is effecient in semi-strong form of EMH as DSEX is not dependent on global and domestic macroeconomic variables in the long run. But, results of the bounds test reveal that the long run cointegrations exist among the variables at 5% level of significance when Shariah index of DSE is dependent on M2, OP, ER, DSEX, SENSEX and DJIMW. The long run coefficients of ARDL results also show that DSEX have a positive and statistically significant long-run effect on DSES in Bangladesh, while M2 and OP are the two negative and significant determinants of DSES in long run. The coefficient of error correction term (-0.86) is negative and significant at 1% significance level suggesting that the long run causalities are also directing from M2, OP, ER, DSEX, SENSEX and DJIMW to DSE Shariah index. Results of the short run dynamics indicate that all of the variables have some significant impact on DSES. Specifically, in the short run, ER is the largest positive determinant of DSE Shariah index followed by DSEX, SENSEX and OP, while M2 and DJIMW are the two negative determinant of DSE Shariah index. The ARDL results also show that a 1% increase in conventional indice contributes to 0.63 % increase in Islamic equity indice in Bangladesh. Besides, a 1% increase in broad money supply and crude oil price contributes to 01.03% and 0.26% decrease respectively in Islamic equity indice. Thus, we can state that DSEX, M2 and OP have some significant impact on DSES in the long run. The results of the estimated ARDL-ECM models reveal that the coefficient of error correction term is negative and statistically significant at the 1% level of significance. It suggests that the long-run causality is also directing from independent variables (M2, OP, ER, DSEX, SENSEX and DJIMW) to DSES. The highly significant ECT implies that 86% of the last month's disequilibrium is corrected this month. The test also reveals that there is positive and significant short run causality running from OP, ER, DSEX and SENSEX to DSES are positive, while the roles of M2 and DJIMW on Islamic indice are negative. Taking into consideration of the results of ARDL bounds test, it is obvious that most of the selected macroeconomic variables do significantly explain the Islamic stock prices of Bangladesh. Since most of the macroeconomic variables do significantly explain the Islamic stock prices of Bangladesh either in the short run or long run or both, it can be concluded that the Islamic stock market is not efficient in semi-strong form. Consequently, investors can earn abnormal profit from the DSES using publicly available information.

This study also investigates the volatility spillover effect using daily return data of DSES and DSEX over the period from 20 January 2017 to 30 June 2017. Employing a bivariate VAR-Granger causality model, we find that there is unidirectional spillovers in returns running from conventional indices to the Islamic indices in Bangladesh. The volatility transmissions across markets are examined using MGARCH-BEKK model and results explore that the volatility is transmitted between these markets. Finally, employing MGARCH-CCC model, we find evidence of very high conditional correlation between Islamic and conventional indices of Bangladesh as the estimated conditional correlation (0.9) is significant at 1% significance level. Therefore, the overall contributions of this study should positively affect the Bangladesh economy, as thorough investigations of conventional and Islamic stock price behavior employing modern econometric techniques give valuable knowledge to investors to allocate their portfolio efficiently and policy makers to regulate existing policies or implement new policies to attract more investments.

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